MIL-C-18640A(WEP)
5 March 1962
Superseding
MIL-C-18640(AER)
15 April 1955

MILITARY SPECIFICATION

CONTROL, RADIO SET C-1763/ARN-21A

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this specification shall provide remote control facilities for Radio Receiver - Transmitter RT-220B/ARN-21 and is designed for console mounting for use in aircraft.

1.2 Classification - The control panel shall consist of the following items:

Items	Type Designation	Appl. Para.
Control, Radio Set	C-1763/ARN-21A	3.4.1

2. APPLICABLE DOCUMENTS

* 2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-W-5088	Wiring; Aircraft, Installation of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-C-6781	Control Panel, Aircraft Equipment, Rack or Console Mounted
MIL-P-997	Plastic Material, Laminated, Thermosetting, Electrical Insulating Sheets, Glass Cloth, Silicone Resin
MIL-P-7788	Plate, Plastic, Lighting

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MIL-E-7894	Electric Power, Aircraft Characteristics of
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-T-18303	Test Procedures, Preproduction and Inspection, for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-F-25173	Fastener, Control Panel, Aircraft Equipment

PUBLICATIONS

Bureau of Aeronautics

SAR-300	Material Changes and Material Bulletins, Preparation of

DRAWINGS

Figure 1	Control, Radio Set (C-1763/ARN-21A), Outline
Figure 2	Control, Radio Set (C-1763/ARN-21A), Schematic

2.2 Availability of Documents:

(1) When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code CDS, 5801 Tabor Avenue, Philadelphia 20, Pa.

3. REQUIREMENTS

* 3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern:

(1) Parts and materials shall conform to Specification MIL-E-5400.

(2) Nonrepairable subassemblies, as outlined in Specification MIL-E-5400, shall be used when practicable. The general size of the subassembly, and the amount of circuitry to be included therein, shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable. (See 6.4)

(3) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable subassemblies without the approval of the procuring activity.

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3.1.1 Nonstandard Parts and Material Approval - Approval for the use of nonstandard parts and materials shall be as outlined in Specification MIL-E-5400.

3.2 Design and Construction:

3.2.1 General - The equipment shall conform with all the applicable requirements of Specification MIL-E-5400 for design, construction and workmanship, unless otherwise specified herein.

3.2.2 Total Weight - The total weight of the equipment, excluding cables, shall be a minimum consistent with good design and shall not exceed 2.25 pounds.

3.2.3 Operating Life -

3.2.3.1 Operational Stability - The equipment shall operate with optimum performance for 500 hours, continuously or intermittently, without the necessity for readjustment of any controls which are inaccessible during flight.

3.2.3.2 Total Operating Life - The equipment shall have a minimum total operating life of 5000 hours with reasonable servicing and replacement of parts. Parts requiring replacement within this interval and the life of such parts shall be specified by the manufacturer.

3.2.4 Cabling and Connections

3.2.4.1 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-E-5400.

3.2.4.2 Interconnection Cabling - The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the equipment.

3.2.5 Control Panel - The control shall meet the requirements of Specification MIL-C-6781, shall have a plastic lighting plate meeting the requirements of Specification MIL-P-7788 and shall be equipped with fasteners meeting the requirements of MIL-F-25173. The lamps shall be connected in parallel with one side grounded to the chassis of the unit and the other side being connected to a separate contact in the electrical connector receptacle of the unit, for connection to an external + 28V input lead. The brilliance of the lamps will be controlled externally and the brilliance control shall not be part of the control unit. The arrangement of controls and all lettering and markings proposed for use on the front panel of the unit shall be approved by the contracting agency prior to fabrication of any complete units.

3.2.6 Interchangeability - The equipment shall meet the interchangeability requirements as defined in Specification MIL-E-5400.

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3.2.7 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of Specification MIL-I-6181.

3.2.8 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in Specification MIL-E-5400.

3.2.9 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307.

3.2.10 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient ($30^{o} \pm 10^{o}C$)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input power voltage	27.5 ± 0.5 V DC

3.2.11 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in Specification MIL-E-5400 for Class I equipment, except as modified herein.

*3.2.12 Warm-up Time - The equipment shall not require any time to warm up prior to operation.

3.2.13 Input Power Requirements - The equipment shall meet all applicable requirements of Specification MIL-E-7894 and shall give specified performance from the following power source with characteristics as defined in Specification MIL-E-7894 having limits as specified hereinafter. The power required shall not exceed the specified limits.

(1) DC Power 28 Volt System

Operating Voltage Limits	24 to 29 volts
Power	2.8 watts

3.2.13.1 Undervoltage Protection - The equipment shall not be damaged by voltages applied below the minimum specified herein and shall automatically resume normal operation when the voltage returns within limits.

3.3 Performance

*3.3.1 General - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard

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and extreme service conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.3.2 Channel Selection - Channel selection shall be made by selecting resistance increments across two series resistor networks connected to two tap switch groups. The resistance from either terminal of each network to ground for any position of the tap switches shall be within 1.0 percent of the nominal value as shown in Figure 1, when measured under standard conditions. When measured under service conditions, the resistances shall be within 1.0 percent of the values measured under the standard conditions.

3.3.3 Function Selection - The function selector shall be a three-position rotary switch provided for the purpose of controlling the operating condition of the RT-220B/ARN-21 by actuating the receive and transmit relays of the RT-220B/ARN-21. In the "OFF" position, neither relay shall be actuated; in the "REC" position, the receive relay only shall be actuated; in the "T/R" position, both receive and transmit relays shall be actuated.

*3.3.4 Identification Output Control - With a 1000-ohm, 2-watt resistor connected between terminals B and J and 40 volts, 400 cycles applied to the same terminals through a 0.22-microfarad capacitor, the Identification Output Control shall provide continuous variation of the audio output power level from a minimum of not more than 5 microwatts to a maximum of not less than 50 milliwatts into either a 300 or a 2000 ohm load (See Fig. 1.)

3.4 Detail Requirements

3.4.1 Radio Set Control C-1763/ARN-21A - The radio set control shall meet the following requirements:

3.4.1.1 Function - The radio set control, when properly interconnected to Radio Receiver - Transmitter RT-220B/ARN-21 and equipments supplying the required power and signal voltages, shall perform the following functions:

(1) Provide channel selection for channels 1 through 126 for RT-220B/ARN-21.
(2) Provide selection of the operating function (Off, Receive, Transmit-Receive) of the RT-220B/ARN-21.
(3) Adjust the Identification Output Level when the equipment is interconnected to an interphone system.

3.4.1.2 Form Factor - Overall size including connector shall not be greater than 5-3/4 inches in width, 3 inches in length, and 4 inches in depth. Controls shall not extend beyond the panel face more than 1-5/16 inches. Outline dimensions shall conform to Figure 2.

3.4.1.3 Weight - The weight of the radio set control shall not exceed 2.25 pounds.

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*3.4.1.4 Contents of Unit - The radio set control shall contain the following circuits.

(1) Channel Selector Circuit - The channel selector circuit shall consist of two precision resistor networks and two coaxially mounted rotary switches actuated by two coaxial knobs, the dials of which are arranged to indicate directly the selected channel number. Both switch assemblies shall be of the shorting or make before break types and be provided with mechanical detents and stops to limit their rotation to the proper number of positions. Dials attached to the selector switches shall be capable of displaying channel indications from 0 to 129, the numbers reading vertically from top to bottom. A suitable mask shall be provided to obscure all the dial numbers except those actually indicating the selected channel. The resistor networks shall employ precision wirewound resistors having a nominal temperature coefficient of not more than ± 30 parts per million per degree centigrade over the temperature range from -55°C to +85°C and having a nominal power rating of not less than 1/2 watt. The resistors shall have the resistance values indicated in Figure 1 and shall be connected in the configuration shown in Figure 1. The resistors shall be mounted on laminated terminal boards per Specification MIL-P-997, type GS-G secured to the frame of the equipment.

(2) Function Selector Circuit - The function selector shall consist of a three-position rotary switch actuated by a wing-type knob. The switch shall be connected as shown in Figure 1. The rotary switch shall be of the shorting or "make-before-break" type and shall employ solid spring silver contact clips and solid coin silver rotor blades.

(3) Identification Output Control Circuit - The identification output control circuit shall consist of a 500 ohms bridged - T type attenuator made up of two fixed resistors and a dual composition variable resistor connected as shown in Figure 1. The fixed resistors shall be of the resistance value shown with a nominal resistance tolerance of not more than ±5% and a nominal power rating of not less than 2 watts. The dual composition variable resistor shall be Allen-Bradley Company type JJ or an approved equivalent and shall have the proper resistance values and resistance variation with shaft rotation to produce approximately linear attenuation in decibels per degree of shaft rotation. The attenuator shall be actuated by a cylindrical knob approximately 0.75 inch in diameter. The cylindrical surface shall be knurled to facilitate gripping.

3.4.1.5 Controls - The radio set control contains the following controls:

(1) Channel Selector "Coarse" Switch - The channel selector "coarse" rotary switch shall be provided with 13 positions and shall be

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connected to the appropriate precision resistor network as shown in Figure 1. The coarse switch shall be actuated by the outer of the two coaxial channel selector knobs. This knob shall be cylindrical in shape and approximately 2 inches in diameter. The cylindrical surface shall be provided with a coarse knurl to facilitate gripping. The dial of the outer knob shall have its maximum counterclockwise position left blank; the remaining 12 positions shall be marked 1 through 12 respectively, the visible numbers to be in increasing order as the knob is rotated clockwise.

(2) Channel Selector "Fine" Switch - The channel selector "fine" rotary switch shall be provided with 10 positions and shall be connected to the appropriate precision resistor network as shown in Figure 1. The fine switch shall be actuated by the inner of the two coaxial channel selector knobs. This knob shall be of the wing-handle type. The dial of the inner knob shall be marked 0 through 9, the visible numbers to be in increasing order as the knob is rotated clockwise.

(3) Function Selector - The function selector shall be actuated by a wing-handle type of operational control mounted on the front panel. Its three positions, connected to the appropriate relays in the RT-220B/ARN-21 transmitter-receiver, shall be labeled "OFF", "REC" and "T/R". This control determines whether the RT-220 will operate in the receive only or transmit and receive modes or be in the off condition.

(4) Identification Output - The identification output level adjustment shall be actuated by a knurled-cylinder type of operational control mounted on the front panel. It shall be labeled "VOL" with a curving arrow to indicate the direction the control must be turned to increase the level. This control sets the level of an Identification signal which can be fed through aircraft interphone circuits.

3.4.1.6 Electrical Connections - Connections to external circuits shall be provided by a 15-pin female contact connector (MS-3102-20-27S) mounted on the rear of the equipment. The connector shall be oriented with the interval key at the top of the equipment. It shall be wired as follows:

Reference Designation	Terminal	Function
J101	A	Receive Control Relay
	B	Identification Signal Input
	C	Transmit Control Relay
	D	Channel Selector - Fine
	E	Channel Selector - Fine
	F	Channel Selector - Coarse
	G	Channel Selector - Coarse
	H	Transmit Control Relay Return

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Reference Designation	Terminal	Function
	I	No Connection
	J	Ground
	K	No Connection
	L	No Connection
	M	Panel Lighting Input
	N	Identification Signal Output

3.4.1.7 Mounting - The radio set control shall be designed for rack and console mounting in accordance with Specification MIL-C-6781. Mounting dimensions shall conform with Figure 2.

4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of Tests - The testing and inspection of aircraft electronic equipment shall be classified as follows:

(1) Preproduction Tests - Preproduction tests are those tests conducted on samples which are representative of the production equipment to be supplied under the contract, for the purpose of determining that the equipment meets all the requirements of the applicable specifications. Preproduction tests shall consist of the following two tests.

(a) Contractor Demonstration Tests - Contractor Demonstration Tests are accomplished under the responsibility of the contractor on a sample, representative of the production equipment, to demonstrate that the equipment complies with all the applicable requirements.

(b) Service Approval Tests - Service Approval Tests are accomplished under the responsibility of the procuring service on a sample representative of the production equipment, to determine that the equipment meets all the requirements of the specification. These tests may be accomplished by having the procuring service supervise the Contractor Demonstration Tests.

(2) Inspection Tests - Inspection Tests are those tests accomplished on equipment submitted for acceptance under the contract. Inspection Tests shall consist of the following:

(a) Individual Test - Individual Tests are those tests conducted on each equipment submitted for acceptance under the contract.

(b) Sampling Tests - Sampling Tests are those tests accomplished on a quantity of equipment as approved by the procuring activity and which have previously passed the Sampling Tests.

(c) Special Tests - Special tests are those tests accomplished on a quantity of equipment as approved by the procuring activity and which have previously passed the Sampling Tests.

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4.2 Preproduction Tests

4.2.1 Contractor Demonstration Tests - The contractor shall conduct tests on one or more sample equipments as necessary to determine that the design of the equipment as proposed by the contractor will meet the requirements of all applicable specifications. Contractor Demonstration Tests shall be conducted in accordance with the approved preproduction test procedure. The data obtained by the contractor in conducting these tests shall be included with the design data submitted with the Service Approval Test sample. The Inspector of Naval Material and the procuring activity shall be advised when tests are to be conducted so that the procuring activity may designate a representative to witness or supervise the test when so desired. When tests are supervised the may decide to omit certain portions of the Service Approval Tests which will duplicate the Contractor Demonstration Tests; however, in conducting Service Approval Tests the procuring activity may elect to repeat any test previously conducted by the contractor, when deemed necessary. Contractors not having laboratory facilities to satisfactorily conduct all tests shall obtain either the services of a commercial testing laboratory or written approval from the procuring activity to omit that portion of the tests which cannot be conducted.

4.2.2 Service Approval Tests - The Service Approval sample shall consist of one equipment representative of the production equipment to be supplied under the contract. This equipment shall be delivered by the contractor to the destination specified by the procuring activity. The sampling procedure outlined in Specification MIL-P-7788 shall apply for plastic lighting plates.

4.2.2.1 Accessory Materials Required - In addition to the complete equipment submitted, the contractor shall submit the accessory material and design data as specified in Specification MIL-E-5400.

4.2.3 Scope of Tests - Preproduction Tests shall include all tests deemed necessary to determine that the model meets all the requirements of this specification and the contract. Preproduction Tests shall include testing in accordance with Specification MIL-T-5422 and Specification MIL-I-6181. The testing of plastic lighting plates shall be in accordance with Specification MIL-P-7788. The design and test data supplied by the contractor shall indicate the equipments physical and electrical characteristics and establish that the equipment meets all the requirements of this specification. The test data shall include data obtained by the contractor in testing to MIL-T-5422.

4.2.4 Preproduction Sample Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all pre-production sample. Prefabrication of any equipments prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample will be returned to the contractor for his use in the fabrication and testing of the equipment to be submitted for acceptance under the contract. The preproduction sample shall not be considered as one of the equipments under the contract, however, it may be reworked by the contractor and submitted for acceptance as a production equipment.

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4.3 Inspection Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the required tests. All inspection and testing shall be under the supervision of the Government Inspector. Contractors not having laboratory facilities satisfactory to the Government Inspector shall engage the services of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all inspection tests required by this specification, and signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during course of manufacture shall in no case be construed as a guarantee of the acceptance of the finished product. Inspection tests shall consist of the following:

(1) Individual Tests - Individual Tests shall be conducted on each equipment submitted for acceptance under the contract.

(2) Sampling Tests - Sampling Tests shall be conducted on a quantity of equipment as specified in 4.3.2.1. Equipments for sampling tests shall be selected by the Government Inspector after the equipment has passed the Individual Tests and has been packaged and packed ready for shipment.

(3) Special Tests - Special Tests shall be conducted on a quantity of equipment as specified in 4.3.3.1. Equipments for Special Tests shall be selected by the Government Inspector after the equipment has passed the Sampling Tests.

4.3.1 Individual Tests - Each equipment shall be examined carefully to determine compliance with the requirements of this specification with respect to material and workmanship. In addition, the equipment shall be given an operational check, with sufficient data recorded to assure satisfactory operation.

4.3.1.1 Scope of Test - As a minimum each equipment shall be tested to determine compliance with the following:

Individual Tests	Applicable Paragraph
I. Tests conducted under standard test conditions and nominal input voltages and frequency.	
1. Check material and workmanship.	
2. Resistor Network Accuracies.	3.3.2
3. Function Selector Control	3.3.3
4. Identification Output Control	3.3.4

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4.3.2 Sampling Tests - Sampling Tests shall be conducted under the close supervision of the Government Inspector and shall include a performance check at ambient laboratory conditions. This Test may also include any of the other preproduction tests the Inspector deems necessary to determine conformance of the equipment with the requirements of this specification. These tests shall include also a careful examination of the way the equipment is packaged and packed to determine conformance with the packaging and packing requirements of this specification or the contract. Sampling tests shall be conducted on complete equipments. When individual units of the equipments are tested individually and shipped then the samples selected for Sampling Tests shall consist of all the individual units necessary to comprise a complete operating equipment.

4.3.2.1 Sampling Test Schedule - Sampling Tests shall be performed on the following quantity of equipment unless otherwise stated in the contract or approved in writing by the procuring service.

Quantity of Equipment	Quantity of Equipment to be Tested
First 10 equipments	1
Next 10 to 100	2
Next 100 to 200	1
Over 200	One for each additional 200

4.3.2.2 Scope of Test - As a minimum, each equipment selected for sampling test shall be tested to determine compliance with the following:

Sampling Tests	Applicable Paragraph
I. Tests conducted Standard Test Conditions And Nominal Input Voltages and Frequency.	
1. Repeat the Individual Tests of Para. 4.3.1.1	

4.3.2.3 Equipment Failure - Should an equipment fail to pass any part of the Sampling Test, the following corrective action shall be taken by the contractor.

(1) Determine the cause of the failure.

(2) Determine if the failure is an isolated case or a design defect.

(3) If a design defect, suggest corrective action and submit to the procuring activity for approval.

(4) Include a test in the Individual Test to check all equipments for this requirement until reasonable assurance can be obtained that the defect has been satisfactorily corrected.

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4.3.3 Special Test - Special Tests shall be conducted under the close supervision of the Government Inspector and shall include a performance check at the environmental conditions specified below using the same test procedures as approved for the Preproduction Test. The Special Test may also include any other tests the Inspector deems necessary to determine conformance of the equipment with the contractual and specification requirements. Special Tests shall be conducted on complete equipments.

4.3.3.1 Special Test Schedule - Special tests shall be conducted on the first and every fifth equipment that successfully passes the Sampling Test.

*4.3.3.2 Scope of Test - As a minimum each equipment selected for Special Test shall be tested to determine compliance with the following:

Special Test	
I. Tests using procedures as approved for the preproduction tests under the following environmental conditions.	
1. High and low temperature	MIL-E-5400
2. Humidity	MIL-E-5400

4.3.3.3 Equipment Failure - Should an equipment fail to pass any part of the Special Test, the following corrective action shall be taken by the contractor:

(1) Determine the cause of the failure.

(2) Determine if the failure is an isolated case of design defect.

(3) If a design defect, suggest corrective action for BuAer approval.

4.4 Test Procedures - The procedures and methods for conducting Preproduction Tests and Inspection Tests shall be prepared by the contractor and sent to the procuring activity for approval. The right is reserved by the procuring activity or the Naval Inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Use MIL-T-18303 as a guide for preparation of Test procedures.

4.5 Pre-Submission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply to the best of his knowledge and belief with all applicable requirements. The design data submitted by the contractor for the preproduction model shall indicate its physical and electrical characteristics and other necessary test data which will establish that the equipment meets all the requirements of this specification.

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4.6 Rejection and Retest - The electronic equipment which has been rejected may be reworked or have parts replaced to correct the defects, and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Inspector. Units rejected after retest shall not be resubmitted without the specific approval of the procuring agency.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be packaged, packed and marked for shipment in accordance with the requirements of Specification MIL-E-17555.

6. NOTES

6.1 Test Values - Normal and limiting values of performance data shall be determined at an input voltage of 27.5 ±0.5 V DC with the radio set control properly interconnected to the RT-220B/ARN-21. These data are to be used in testing the equipment for compliance with minimum acceptable standards of performance.

6.2 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specification are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specification. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.3 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

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6.4 As a general rule non-repairable subassemblies should be encapsulated or hermetically-sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high voltage power supplies), the non-repairable subassembly should evidence a Mean-time-to-failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours.

6.5 An asterisk preceding a paragraph number denotes a change in that paragraph from the basic specification.

NOTICE - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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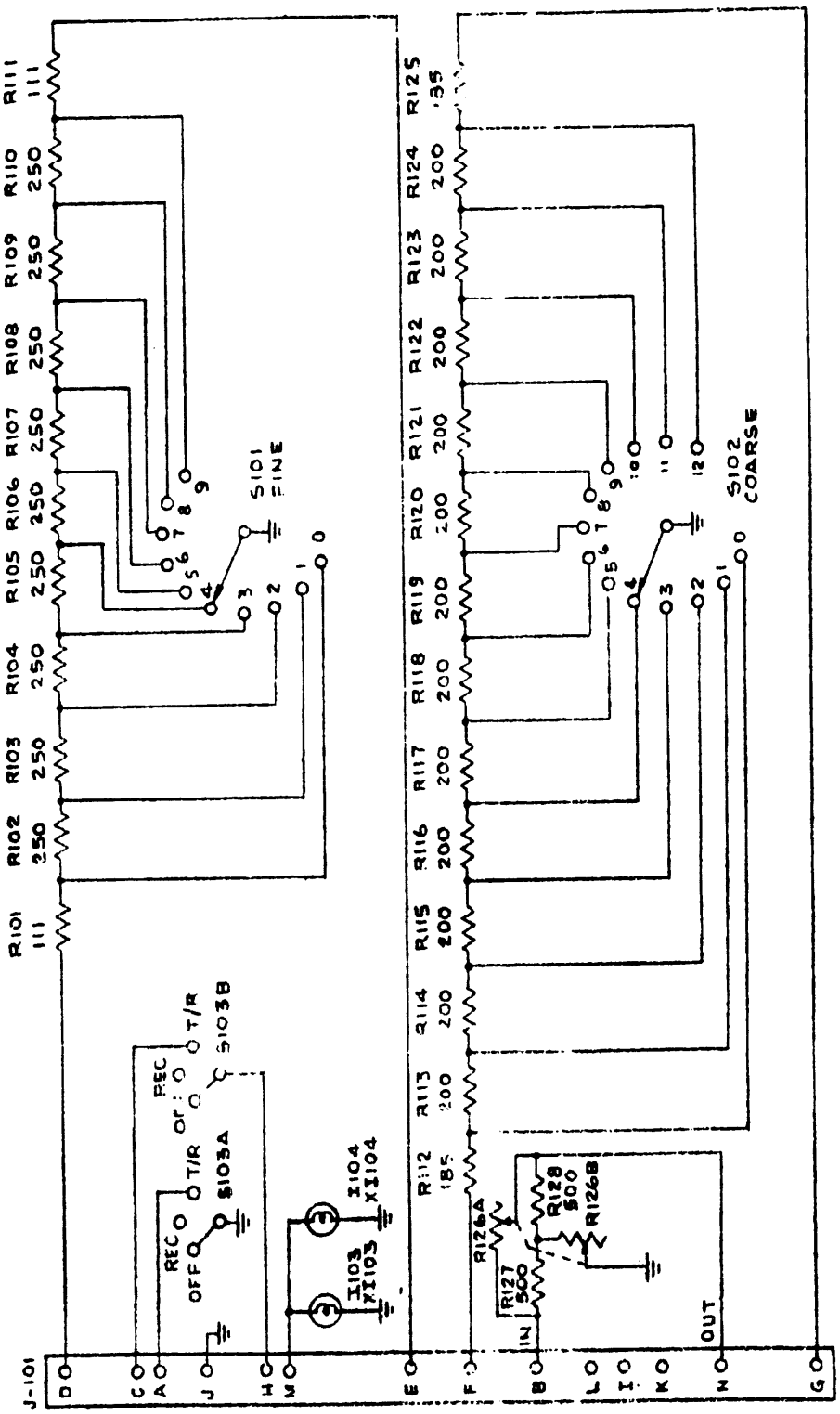


FIGURE 1 SCHEMATIC DIAGRAM - C-1763/ARN-21A

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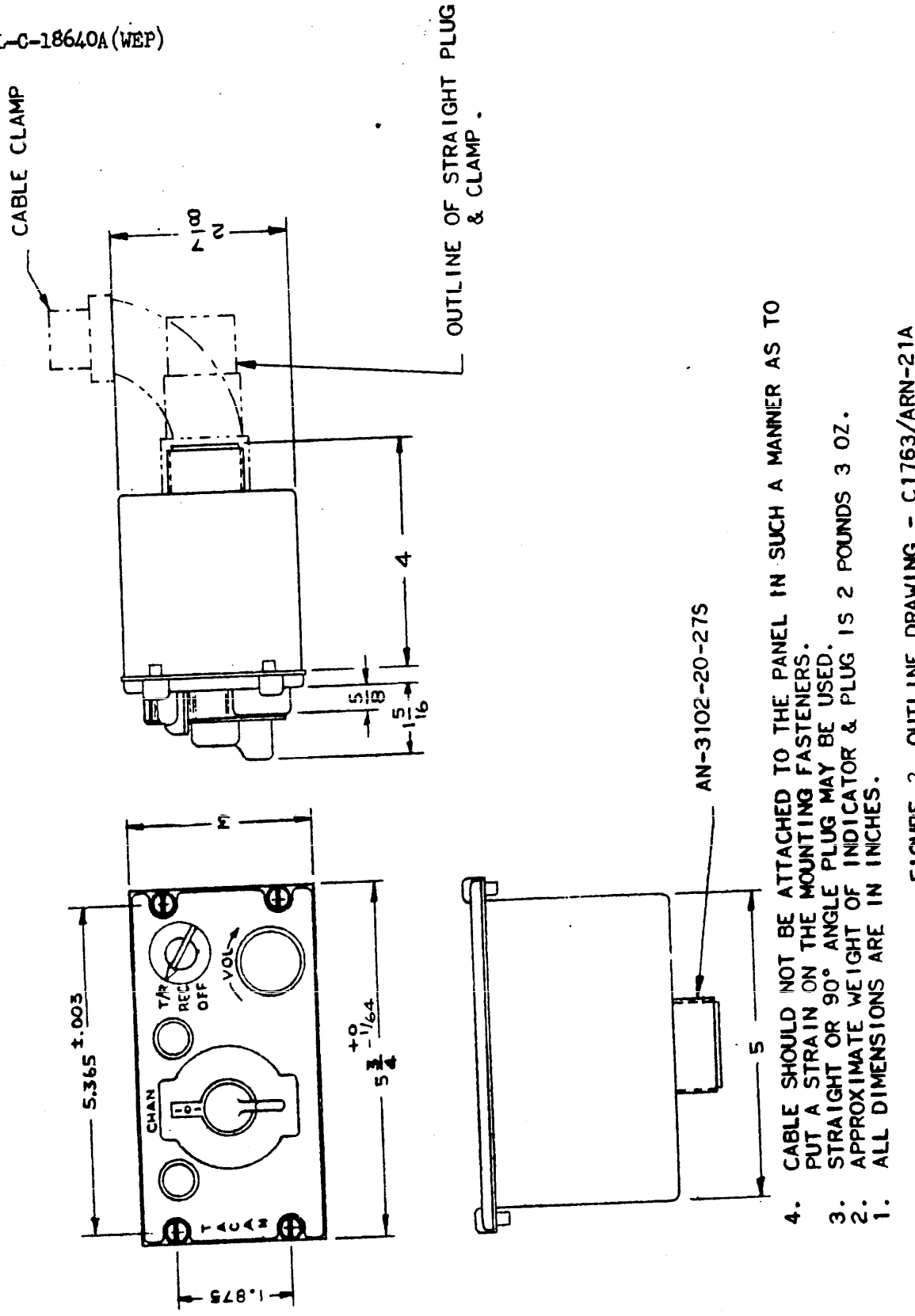


FIGURE 2 OUTLINE DRAWING - C1763/ARN-21A

4. CABLE SHOULD NOT BE ATTACHED TO THE PANEL IN SUCH A MANNER AS TO PUT A STRAIN ON THE MOUNTING FASTENERS.
3. STRAIGHT OR 90° ANGLE PLUG MAY BE USED.
2. APPROXIMATE WEIGHT OF INDICATOR & PLUG IS 2 POUNDS 3 OZ.
1. ALL DIMENSIONS ARE IN INCHES.

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